Errata for Amy, Ross and Wesley,

"A Sound and Complete Equational Theory for 3-Qubit Toffoli-Hadamard Circuits"
(QPL 2024, EPTCS 406, arXiv:2407.11152v3)

Found by the Agda formalisation in `Examples/Groups/CCX+HH-3qubit-opus`

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Page numbers are those of the arXiv version. One relation is missing from the presentation of Theorem 5.2, which makes that theorem false as stated and leaves a gap in the proof of Theorem 6.3. Seven printed equations are false, two of them in four places each. One paragraph contradicts the paper's own formulas, one proof step is incomplete, and there are a few small slips.

Every false equation below is refuted in `Errata.agda` by one entry at which its two sides differ, and every correction is checked there or elsewhere in the development, as noted. The script `gen/errata.py` repeats every check in exact rational arithmetic, and checks the schemas (63) and (66) at every choice of indices. File names are relative to `Examples/Groups/CCX+HH-3qubit-opus/`.

Conventions. These are the paper's own (Section 2):

- $M_0 = M \otimes I \otimes I$, so qubit 0 is the most significant and $|x_0x_1x_2\rangle$ is basis state $4x_0+2x_1+x_2$.
- $CM_{j,k}$ applies M to qubit k when qubit j is $|1\rangle$, and $CCM_{j,k}$ applies M to the third qubit when qubits j and k are both $|1\rangle$.
- Products, written $\cdot$ or $\circ$, are matrix products, so the rightmost factor acts first.
- Matrix entries are given as (row, column), counted from 0.

1 The relation r_K is missing from R_1 (Theorem 5.2)

Section 5.5 (p. 12) introduces the true relation

$$(r_K) \qquad K_{1,2} \approx K_{[4,5,6,7]} \cdot X_0 \cdot K_{[4,5,6,7]} \cdot X_0,$$

but $R_1 = R_0 \cup R_8^4$ (Section 5.6, p. 12) consists of Figures 4 and 7 only, and neither contains it. Figure 4 never mentions $K_{[0,1,2,3]}$ or CCZ, and Figure 7 never mentions $K_{1,2}$, so no relation of R_1 involves both $K_{1,2}$ and $K_{[0,1,2,3]}$.

r_K does not follow from R_1. Here R_1 means Figures 4 and 7 with the corrections of Section 2 below. Let J be the 8×8 all-ones matrix and $R = I - J/4$. Then $R^2 = I$, and R commutes with every permutation matrix. Read each gate g of Figure 4 as RgR, and read $K_{[0,1,2,3]}$ and CCZ as themselves:

- Every relation of Figure 4 holds, since it is conjugated by R.
- Every relation of Figure 7 holds unchanged. Its gates are $K_{[0,1,2,3]}$, CCZ and circuits of permutation gates, and R commutes with those circuits.
- r_K fails: its left side has entry $(0,0)$ equal to $-1/4$, its right side $1/2$.

So $\langle \Sigma_1 \mid R_1 \rangle$ is not $O(8, \mathbb{D})$, and Theorem 5.2 is false as stated.

Correction. With r_K added, $R_1^+ = R_1 \cup \{r_K\}$ presents $O(8,\mathbb{D})$; this is `theorem-5-2` in `Theorems.agda`, and the countermodel is `r_K-independent` (`Independence.agda`). There r_K is stated as $K_{1,2} \approx X_0\, K_{[0,1,2,3]}\, X_0\, K_{[0,1,2,3]}$. That is the same relation, because the printed right-hand side is $K_{[4,5,6,7]}K_{[0,1,2,3]}$, this one is $K_{[0,1,2,3]}K_{[4,5,6,7]}$, and (96) equates the two.

Effect on Theorem 6.3. Its proof (p. 14) uses Theorem 5.2 ("$O(8,\mathbb{D}) \cong \langle \Sigma_1, R_1 \rangle$ with $R_1 \subseteq R_2$"). With r_K added, $R_2^+ = R_2 \cup \{r_K\}$ presents TofH(3): this is `theorem-6-3`. We do not know whether r_K follows from R_2, that is, whether Figure 8 makes up for the missing relation. The countermodel above does not extend to H_2 directly: the only 8×8 matrix that satisfies (102)–(107) with H_2 read as it is 0, which fails (108).

Counts. The presentations have 66 and 73 relations, not the 65 of the Introduction (p. 1) and the Conclusion (p. 14).

2 False equations

Each item gives the printed equation, its correction, and an entry at which the two printed sides differ; the name in brackets is the refutation in `Errata.agda`. Write K for $K_{[0,1,2,3]}$.

(33), Figure 4, p. 8 [`r033-false`]

Printed: $CCX_{0,1} \approx \sigma_{1,2} \cdot CCX_{0,2} \cdot \sigma_{0,2}$
Correct: $CCX_{0,1} \approx \sigma_{1,2} \cdot CCX_{0,2} \cdot \sigma_{1,2}$

Entry $(1,1)$: 1 against 0. As printed, R_0 is unsound. The correction is relation `r33` of `Syntactics.agda`, checked by `Soundness.agda`.

Section 5.5, p. 11 [`x67-false`]

Printed: $CCX_{1,2} = X_{[6,7]}$
Correct: $CCX_{0,1} = X_{[6,7]}$

Entry $(3,3)$: 0 against 1. The paper uses this to replace $X_{[6,7]}$ in R_8^3 by $CCX_{1,2}$. In fact $CCX_{1,2} = X_{[3,7]}$ (`ccx12`), and $X_{[6,7]}$ is $CCX_{0,1}$, as the paper itself says on p. 9. So $X_{[6,7]}$ in Figure 7, in (90) and (93), must be read as $CCX_{0,1}$. Section 5.6 lists circuits only up to $X_{[5,6]}$. $CCX_{0,1} = X_{[6,7]}$ is `StdMatrices.mT6`.

(r_{CX}), Section 5.5, p. 11 [`rCX-false`]

Printed: $CX_{0,1} \approx X_{[2,6]} \cdot X_{[3,7]}$
Correct: $CX_{0,1} \approx X_{[4,6]} \cdot X_{[5,7]}$

Entry $(2,2)$: 1 against 0. $X_{[2,6]}X_{[3,7]}$ is $CX_{1,0}$ (`cx10`), and the correction is `rCX`. r_{CX} is a step of the Tietze sequence only and does not reach Figure 7.

The alias of $(-1)_{[0]}$, Section 5.6, p. 12 [`m0-false`]

Printed: $(-1)_{[0]} = X_0X_1\, CCZ\, X_1X_0$
Correct: $(-1)_{[0]} = X_0X_1X_2\, CCZ\, X_2X_1X_0$

Entry $(0,0)$: -1 against 1. The printed word is $(-1)_{[1]}$ (`m1`). The corrected alias is `m0` in `Syntactics.agda`, checked by `StdMatrices.mm0`.

(97), Figure 7, p. 12 [`r097-false`]

Printed: $X_{[0,1]}\,K \approx K\,X_{[0,1]}\,(-1)_{[1]}\,(-1)_{[3]}$
Correct: $X_{[0,1]}\,K \approx K\,X_{[1,3]}\,(-1)_{[1]}\,(-1)_{[3]}$

Entry $(1,1)$: $1/2$ against $-1/2$. The correction is Li, Ross and Selinger's (4a). The same misprint is in (78) (Figure 6, p. 11) and (125) (Figure 10, p. 41), which are (97) on the two-level generators, and in the schema (63) (Figure 5, p. 9), where the correct right-hand side is $K_{[a,b,c,d]}\,X_{[b,d]}\,(-1)_{[b]}\,(-1)_{[d]}$. As printed, (63) fails at all 70 choices of $a < b < c < d < 8$, and the corrected schema holds at all of them (`gen/errata.py`). The correction is relation `r97`, checked by `Soundness.agda`.

(100), Figure 7, p. 12 [`r100-false`]

Printed: $K\,K_{[1,3,4,5]} \approx K_{[1,3,4,5]}\,K$
Correct: $K\,K_{[1,3,4,5]} \approx K_{[2,3,4,5]}\,K_{[0,1,2,4]}$

Entry $(0,3)$: 0 against $1/2$. The correction is Li, Ross and Selinger's (5a). The same misprint is in (81) (Figure 6) and (128) (Figure 10), and in the schema (66) (Figure 5), whose correct form is $K_{[a,b,c,d]}\,K_{[b,d,e,f]} \approx K_{[c,d,e,f]}\,K_{[a,b,c,e]}$. As printed, (66) fails at all 28 choices of six indices, and the corrected schema holds at all of them. The correction is relation `r100`.

The CCZ decomposition, Section 5.7, p. 13 [`ccz-false`]

Printed: $CCZ = K_{1,2} \circ CZ_{1,2} \circ X_0 \circ K \circ X_0 \circ CZ_{1,2} \circ K \circ X_{[5,6]}$
Correct: $CCZ = K_{1,2} \circ CZ_{1,2} \circ X_0 \circ K \circ X_0 \circ CZ_{1,2} \circ K_{1,2} \circ X_{[5,6]}$

Entry $(4,4)$: 1 against $1/2$. The printed equation is false in either reading of $\circ$. The correction replaces the second K by $K_{1,2}$ (`ccz`); by r_K, that factor is $K_{[0,1,2,3]}\,K_{[4,5,6,7]}$. The other decomposition, $K = (K_{1,2} \circ CCZ)^3 \circ X_{[5,6]}$, is right as printed (`nk`). Neither presentation uses these decompositions.

3 The control states of $K_{[0,1,2,3]}$ (Section 5.5, pp. 11–12)

The paper says that $K_{[0,1,2,3]}$ is a $K_{1,2}$ gate applied when qubit 0 is in state $|1\rangle$, and $X_0 \circ K_{[0,1,2,3]} \circ X_0$ one applied when qubit 0 is in state $|0\rangle$. It is the other way round: $K_{[0,1,2,3]}$ acts on states 0–3, where qubit 0 is $|0\rangle$. The paper says so itself at the end of Section 5.6, where $K_{[0,1,2,3]}$ is "a negatively controlled K gate", and on p. 9, where $K_{[4,5,6,7]}$, which is $X_0\,K_{[0,1,2,3]}\,X_0$, is "a controlled K-gate". The relation r_K itself is right.

4 A gap in the proof of Lemma 6.1 (pp. 13–14)

The inductive step moves H_2 past the first letter x of w "by one of Relations (102), (103), (104), (105), (106) and (107)". Those relations cover only six letters of Σ_1: X_0, $CX_{0,1}$, $CCX_{1,2}$, CCZ, $K_{1,2}$ and $K_{[0,1,2,3]}$. Σ_1 has 25 letters. For the other 19 (X_1, Z_0, $CZ_{0,1}$, $\sigma_{0,1}$, …), x must first be rewritten by its defining relation, (1)–(19) of Figure 4. These define each derived gate of Σ_D in terms of earlier ones, and H_2 is then moved past the result letter by letter: an induction on the order of the definitions. The lemma is true: `Push.agda` proves it this way.

5 Small slips

- **p. 3, the definition of $CCM_{j,k}$:** $l \in \{1,2,3\} \setminus \{j,k\}$ should be $l \in \{0,1,2\} \setminus \{j,k\}$.
- **p. 13, Section 5.7:** "the set $\Sigma_D \cup \{CCZ\}$ is minimal, whereas the set $\Sigma_D \cup \{K_{[0,1,2,3]}\}$ is not". Σ_D itself is not minimal, since it contains the 19 derived gates. Presumably Σ_0

is meant: $\Sigma_0 \cup \{CCZ\}$ is Σ_Z of Theorem 5.3, and in $\Sigma_0 \cup \{K_{[0,1,2,3]}\}$ the gate $K_{1,2}$ is redundant, by r_K.

- **p. 14, proof of Lemma 6.1:** in "Then $H_2 \cdot u \sim_{R_2} u' \cdot w' \cdot H_2$ by the transitivity of $(\sim_{R_2})$", $H_2 \cdot u$ should be $H_2 \cdot w$.
- **p. 31, Appendix C:** "$CX_{1,2}$ is a reflection about the normal vector $|-\rangle \otimes |1\rangle \otimes |1\rangle$". That reflection is $CCX_{1,2}$. $CX_{1,2}$ is not a reflection: its -1-eigenspace has dimension 2.
- **p. 41, Figure 10:** ρ ends in an extra $X_{[0,4]}$ that it does not have in Figures 5–7. This is harmless, because (129) holds with either ρ.

Checking

`Errata.agda` typechecks in about half a minute given the interfaces of this directory (Agda 2.8.0, standard-library 2.4, `--safe --without-K`). `gen/errata.py` needs only Python 3: run `python3 errata.py` in `gen/`.

An independent formalisation of the same paper, in `Examples/Groups/CCX+HH-3qubit/`, records the same corrections for Sections 1–3, except r_{CX} and the CCZ decomposition. Those two, and Sections 4 and 5, are new here.